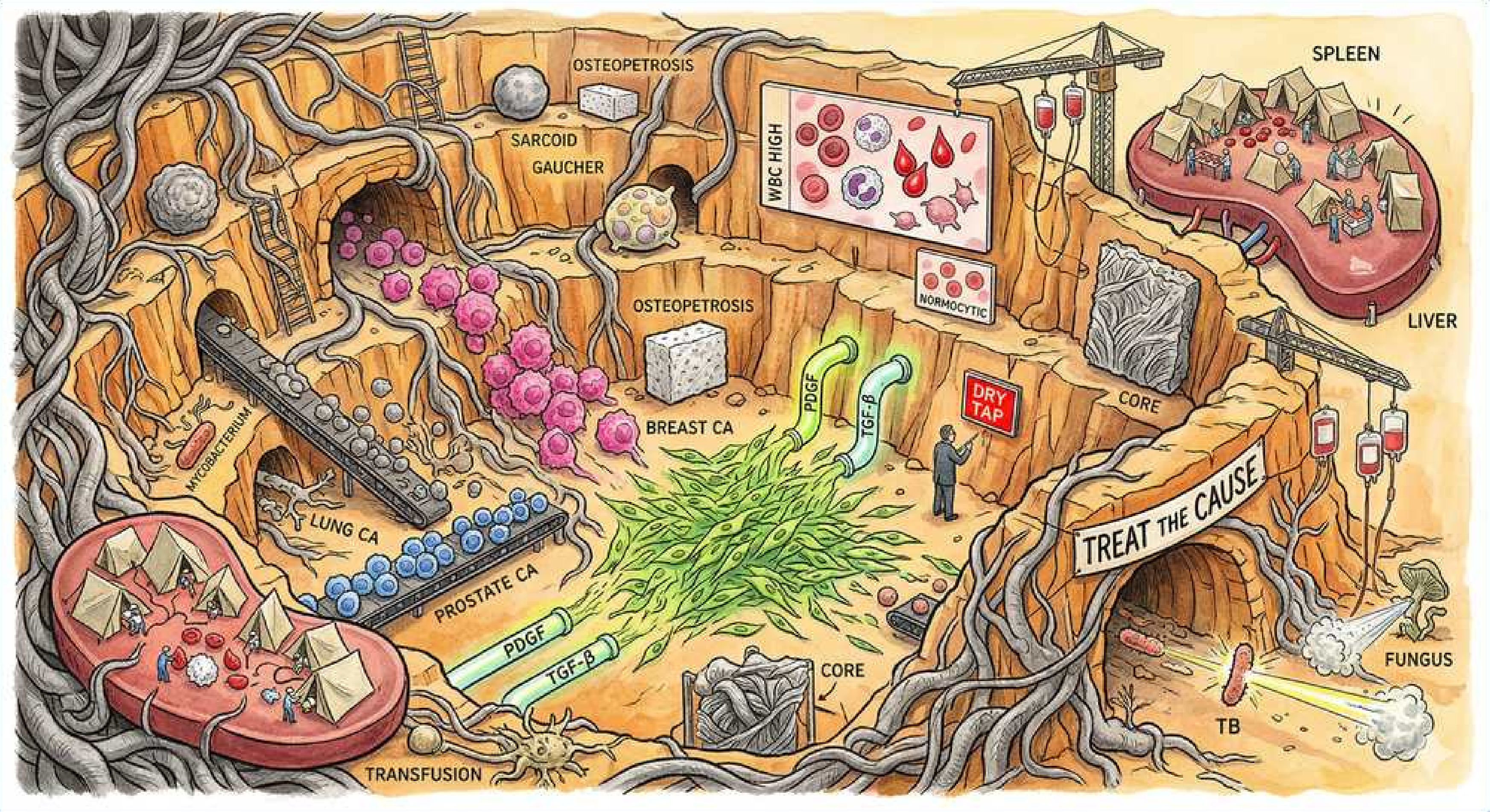
Bone Marrow Failure — The Fibrotic Quarry (Myelophthisic Anemias)



1. Fibrotic quarry

WHAT YOU SEE

Stone quarry walls choked with grassy fibrosis and a worker

WHAT IT ENCODES

Myelophthisic anemia means the marrow space is physically invaded or replaced — by fibrosis, tumor, granuloma, or storage material — crowding out normal hematopoiesis and producing cytopenias. The hallmark is that the marrow is NOT empty (as in aplastic anemia) but packed with abnormal tissue, which forces immature precursors out into the blood.

BOARD TRAP

Don't confuse the INFILTRATED/replaced marrow of myelophthisis with the EMPTY, hypocellular, fatty marrow of aplastic anemia — the smear is the giveaway (leukoerythroblastic + teardrops in myelophthisis vs. pure pancytopenia without immature forms in AA).

2. Leukoerythroblastic smear

WHAT YOU SEE

Wall monitor showing WBC HIGH with mixed immature cells

WHAT IT ENCODES

The defining peripheral-blood picture is a leukoerythroblastic smear: immature myeloid precursors (myelocytes, metamyelocytes — a left shift) appear alongside nucleated red cells (normoblasts). The disrupted marrow architecture lets precursors escape prematurely into circulation. This triad — left-shifted granulocytes + nucleated RBCs + teardrop cells — should prompt a marrow biopsy.

BOARD TRAP

A leukoerythroblastic smear is not specific for malignancy alone — severe sepsis, massive hemorrhage, and severe hemolysis can transiently mimic it, but persistent leukoerythroblastosis with teardrops strongly implies marrow infiltration/fibrosis.

3. Teardrop cells

WHAT YOU SEE

Teardrop-shaped red cells (dacrocytes) on the upper screen

WHAT IT ENCODES

Teardrop cells (dacrocytes) are red cells deformed as they are squeezed through a fibrotic/infiltrated marrow and the splenic cords during extramedullary hematopoiesis. They are the single most classic morphologic clue to marrow fibrosis or infiltration, and are a hallmark of primary myelofibrosis.

BOARD TRAP

Teardrop cells point to marrow stress/infiltration but are not pathognomonic for any one cause — pair them with the clinical context (massive splenomegaly + JAK2 → myelofibrosis; bone pain + known primary → metastatic carcinoma).

4. Nucleated RBCs

WHAT YOU SEE

Red cells with nuclei on the lower display screen

WHAT IT ENCODES

Nucleated RBCs (normoblasts) in the peripheral blood of an adult are abnormal and signal marrow stress: either precursors are being pushed out by infiltration (myelophthisis) or there is intense erythropoietic drive/extramedullary hematopoiesis. Combined with immature myeloid forms they complete the leukoerythroblastic picture.

BOARD TRAP

In an adult, any circulating nucleated RBCs warrant explanation — together with teardrops and a left shift they signal marrow infiltration; in isolation they may reflect severe hemolysis, hypoxia, or post-splenectomy state.

5. Dry tap

WHAT YOU SEE

Red DRY TAP sign in the quarry

WHAT IT ENCODES

A 'dry tap' — failure to aspirate marrow — occurs when fibrosis or packed tumor makes the marrow inaspirable. This mandates a core trephine BIOPSY (with reticulin/collagen staining) rather than aspiration to demonstrate the fibrosis or infiltrate and reach a diagnosis.

BOARD TRAP

A dry tap should never be reported as 'aplastic/hypocellular' — it usually means the marrow is too FIBROTIC or PACKED to aspirate. Always proceed to core biopsy; an aspirate alone can completely miss myelofibrosis or metastatic carcinoma.

6. Breast cancer mets

WHAT YOU SEE

Cluster of pink BREAST CA tumor cells

WHAT IT ENCODES

Metastatic carcinoma is the most common cause of myelophthisic anemia in adults. The classic marrow-seeking solid tumors are breast, prostate, and lung (also gastric, thyroid, renal). In a patient with a known primary plus new cytopenias, teardrops, and a leukoerythroblastic smear, marrow metastasis tops the differential.

BOARD TRAP

New unexplained cytopenias plus bone pain in a patient with breast/prostate/lung cancer = think marrow metastasis (myelophthisis), not chemotherapy myelosuppression alone — biopsy confirms.

7. Prostate cancer

WHAT YOU SEE

Blue PROSTATE CA cells along a conveyor

WHAT IT ENCODES

Prostate carcinoma classically metastasizes to bone with OSTEOBLASTIC (sclerotic) lesions and can infiltrate the marrow to cause myelophthisis. Look for the combination of a very high alkaline phosphatase/PSA, sclerotic bone lesions, and a leukoerythroblastic smear.

BOARD TRAP

Prostate (and breast) bone mets are typically osteoBLASTIC/sclerotic — multiple myeloma's are osteoLYTIC. Don't reflexively call sclerotic marrow-infiltrating lesions myeloma.

8. Lung cancer

WHAT YOU SEE

LUNG CA tumor mass at left

WHAT IT ENCODES

Lung cancer (especially small cell) is another common solid tumor that infiltrates the marrow, producing cytopenias and a leukoerythroblastic blood picture. It may also cause cytopenias via paraneoplastic mechanisms, so marrow biopsy distinguishes direct infiltration from other causes.

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9. Myelofibrosis / PDGF

WHAT YOU SEE

PDGF and TGF-β pathways feeding the grassy fibrosis

WHAT IT ENCODES

Primary myelofibrosis is a myeloproliferative neoplasm in which clonal megakaryocytes/monocytes release fibrogenic cytokines — PDGF and TGF-β — that stimulate reactive (polyclonal) fibroblast collagen/reticulin deposition. Drivers are JAK2 V617F (~50-60%), CALR, or MPL mutations. Clinically: massive splenomegaly (extramedullary hematopoiesis), teardrop cells, leukoerythroblastosis, and a dry tap. JAK inhibitors (ruxolitinib) reduce spleen size and symptoms; allogeneic transplant is the only cure.

BOARD TRAP

The fibroblasts in myelofibrosis are a REACTIVE, polyclonal response to cytokines from the malignant clone — the fibroblasts themselves are not neoplastic. Also: massive splenomegaly + teardrops + JAK2 favors primary myelofibrosis over metastatic carcinoma.

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10. Granulomatous (TB/fungus/sarcoid)

WHAT YOU SEE

TB rods and FUNGUS at lower right, SARCOID upper left

WHAT IT ENCODES

Infiltrative infections and granulomatous disease can replace marrow: disseminated tuberculosis, disseminated fungal infection (histoplasmosis), and sarcoidosis form marrow granulomas that displace hematopoiesis. Marrow biopsy with AFB/fungal stains and culture is key in a patient with fever, weight loss, and cytopenias.

BOARD TRAP

In an immunocompromised or febrile patient with pancytopenia, granulomatous marrow infiltration (disseminated TB/histoplasmosis) is a treatable, easily-missed cause — always stain and culture the marrow, don't just label it idiopathic.

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11. Storage / Gaucher

WHAT YOU SEE

GAUCHER and SARCOID labels with stored-cell boulders

WHAT IT ENCODES

Storage diseases infiltrate marrow with lipid-laden macrophages — Gaucher disease (glucocerebrosidase deficiency) produces 'crumpled tissue paper' Gaucher cells that crowd out hematopoiesis, causing anemia, thrombocytopenia, and hepatosplenomegaly. Enzyme replacement therapy is the treatment.

BOARD TRAP

Gaucher cells (wrinkled tissue-paper cytoplasm) are macrophages stuffed with glucocerebroside — distinguish from the foamy 'sea-blue' or Niemann-Pick storage cells; the clue is splenomegaly + cytopenias + bone crises in a young adult.

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12. Osteopetrosis

WHAT YOU SEE

OSTEOPETROSIS dense-bone labels in the upper quarry

WHAT IT ENCODES

Osteopetrosis results from defective osteoclasts (e.g., carbonic anhydrase II or chloride-channel defects) that cannot resorb bone, so dense sclerotic bone progressively obliterates the marrow cavity, causing marrow failure with anemia and thrombocytopenia. The malignant infantile form is treated by hematopoietic stem cell transplant (which restores osteoclasts).

BOARD TRAP

Osteopetrosis bone is DENSE but BRITTLE — paradoxically prone to fractures despite high radiographic density. The marrow failure is from physical obliteration of the cavity, and transplant works because osteoclasts derive from the hematopoietic lineage.

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13. Spleen/liver EMH

WHAT YOU SEE

SPLEEN and LIVER organs at upper right with cell deposits

WHAT IT ENCODES

When the marrow can no longer support hematopoiesis, blood-cell production reverts to fetal sites — the spleen and liver — producing extramedullary hematopoiesis (EMH) with resultant hepatosplenomegaly (often massive splenomegaly in myelofibrosis). EMH also contributes to teardrop-cell formation as cells traverse the spleen.

BOARD TRAP

Massive splenomegaly with teardrop cells and a dry tap is the classic primary-myelofibrosis picture — but EMH can occur in any chronic marrow-failure/infiltration state, so it's a clue to chronicity, not a specific diagnosis.

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14. Treat the cause

WHAT YOU SEE

Tunnel banner TREAT THE CAUSE, plus TRANSFUSION at lower left

WHAT IT ENCODES

Management of myelophthisic anemia is directed at the underlying process: treat the cancer (systemic therapy/radiation for metastatic carcinoma), the infection (anti-TB/antifungal), the storage disease (enzyme replacement), or the MPN (JAK inhibitors, allogeneic transplant for myelofibrosis). Transfusions and supportive care bridge while the cause is addressed.

BOARD TRAP

There is no single 'myelophthisis treatment' — the cause dictates therapy, so the diagnostic core biopsy (with stains, cytogenetics, JAK2/CALR/MPL testing, and search for a primary) must come before management.
